
Ayurveda Based Food Recommendation System.

Final Report

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Abstract—This report presents the development of an Ayurveda-based food recommendation system promoting healthier and more balanced eating habits for the users’ specific body types also known as doshas (Vata, Pitta, and Kapha). Utilizing over 28,000 validated recipes from Food.com and Allrecipes.com, the system employs Natural Language Processing (NLP) to extract and analyze ingredients, scoring them based on Ayurveda dietary principles. The recommendation engine leverages a K-Nearest Neighbors (KNN) algorithm, optimized for nutritional content and dosha compatibility, to provide personalized meal suggestions. Additionally, a web platform was developed to allow users find their doshas through a comprehensive questionnaire and deliver tailored recipe recommendations and weekly meal plans just for them. This holistic approach integrates traditional Ayurvedic knowledge with modern data science techniques to enhance user well-being through diet.

I. INTRODUCTION

Ayurveda is a traditional system of medicine originating from India, with over 5,000 years of history. Its name comes from Sanskrit and means “science of life.” It is based on the balance of the three doshas (vata, pitta, and kapha), which represent different vital energies. Ayurveda uses a combination of diet, herbs, exercise, meditation, and detoxification practices to promote physical, mental, and spiritual health. Its holistic approach seeks to treat not only the symptoms of diseases but also their underlying causes, thus fostering overall well-being. The project we have initiated focuses on building a food recommendation system based on the principles of Ayurveda. The main objective of this project is to create a tool that can guide users to adopt healthier and more balanced eating habits according to their specific dosha (vata, pitta, or kapha). This system allows users to understand certain personal characteristics, such as body type, health status, and lifestyle, to provide personalized food recommendations based on their primary dosha. Additionally, it has the potential to advise on efficient food combinations for each dosha.

To build this system we extracted and collected over 350,000 food recipes from two sites dedicated to collabo-

rative food publication and preparation: food.com and all-recipes.com. After establishing a data cleaning and validation system, we obtained over 28,000 valid recipes to work on and to use as recommendation for the end user of this system. This valid recipes considers complete data of nutritional information, preparation time and categorization. The nutritional information was used to build two main scores: a health score, based on indicators provided by the World Health Organization (WHO) and a second score, to measure the recipe suitability on each dosha type. This second score was obtained using a database of Ayurvedic ingredients and specific dietary recommendations for each dosha. With the help of NLP techniques, we extracted ingredients from each recipe using their preparation text. Then an ingredient analysis was carried out to calculate a score that measures how many ingredients recommended for an specific dosha are present in a given preparation. Using this calculated score, It was possible to obtain an approximate classification of each preparation in each dosha type, also obtaining recipes that did not contain any Ayurvedic ingredients.

The second part of this work involved the construction of a platform that could be used by users to be classified into a certain dosha. For this, a web service was developed that includes a questionnaire designed to collect relevant data on lifestyle, diet, body composition, and other important aspects, thereby determining an approximate percentage of the user in each dosha. Finally, the user receives recommended recipes according to their type. Each recommended recipe has been adjusted using a KNN algorithm, which can recommend similar recipes. The algorithm has been optimized using the main nutrients of the recipe, preparation time, and the recipe’s score in each dosha. This algorithm is configured to penalize recipes that have a different score in a dosha, ensuring that the user mainly receives recipes that are healthy according to their type. Finally, the user can plan a week of meals using the recipes with the best nutritional scores. The following sections of this report detail each of the described stages.

II. METHODOLOGY

A. Data Extraction

To begin, we required recipe data to offer recipes to users based on their Ayurvedic dosha type. First, we used Food.com recipes data obtained from Kaggle, a data science competition platform and online community for data scientists and machine learning. The dataset includes all the columns necessary for developing the recipe recommendation system. To build a more reliable recommendation system, we needed entire recipes, thus we deleted null values throughout the data analysis and cleaning stage because the dataset was highly unclean. We used a thorough cleaning method that was suitable for our system.

We also opted to use a web scraping technique to make our system more resilient and thorough, as it aggregates data from an additional website, providing data diversity. So we scraped the recipe data from the allrecipes.com site. To do this, we created a web crawler that crawls the website and collects only data that is intended for public viewing. To ensure that our web crawler functions legally and ethically, we used a robots.txt file verification. This file works as a standard used by websites to indicate to visiting web crawlers and other web robots which portions of the website they are allowed to visit. This also enables the site owner to manage external users actions and prevent overloading the web server and avoids indexing pages that are not intended for public access [1]. During the recipe scraping stage of this project, we used open source Python libraries such as BeautifulSoup, Selenium, and urllib.

Urllib is a package that includes many modules, such as urllib.request for opening and reading URLs and urllib.robotparser for parsing robots.txt. Selenium is a library used to perform actions on websites, navigate across pages, and retrieve data from them. Beautiful Soup is a Python module used to parse HTML and XML texts. It constructs a parse tree from these documents, facilitating data extraction from HTML and making it a crucial instrument for online scraping.

Using all of these libraries, we were able to scrape partial data from the allrecipes.com website and create a new dataset, which we then combined with the food.com dataset obtained through the Kaggle platform. We mention partial data since the recipe links we scraped have the following format: 'https://www.allrecipes.com/recipe/'. We ignore any other recipe URLs (for example, '-recipe-').

To conclude, the entire extraction procedure allowed us to mix recipes from two distinct websites, which led to better food recommendations to the application users

B. Building Recommendation System

After some research about Ayurveda, the 2 possible approaches to build the recommendation system were identified. First, recommendation of the recipes based on nutritional data, such as calories, fat content, cholesterol

content, sugar content, and others. Second, suggestion of the recipes based on ingredients used in those recipes. We decided to implement both techniques in the following way: initially, we recommend the top 20 recipes based on Ayurvedic dosha type, and then the user can view related recipes found using the KNN algorithm.

1) *Ingredient based food recommendation:* The first strategy involves recommending meals based on the ingredients they contain. Ayurveda advises that persons be classified into three dosha types: Vata, Pitta, and Kapha. We performed research and prepared an Excel spreadsheet for three dosha kinds, including foods that certain dosha types can consume and those that they should avoid. The entire list for each dosha type is presented in Dosha type.xlsx file.

During this step, it is necessary to have an ingredient list for each recipe in the correct format. However, the second dataset that was obtained by scraping allrecipes.com has many unnecessary data inside a recipe_ingredient column that should be filtered. There are a lot of Natural Language Processing tools available in Python that can be used for this task. However, due to its more accurate semantic analysis, it was decided to use the NLP spaCy model [2]. In this task, we used "en_core_web_sm" which is a small English pipeline provided by spaCy and trained on written web text. "en" here represents the language of the text, that is English. In general, one of the core features of the spaCy model is the tokenizer. Tokenization is the process of splitting a text into individual units called tokens. In this case, tokens are words, punctuation marks, and numbers. So, tokenization helps in breaking down the recipeingredient column text into smaller manageable pieces for further processing. spaCy tokenizer is rulebased and highly efficient [3]. The default tokenizer was working partially correctly. However, additional adjustments had to be made for better extraction results. So, by default, the tokenizer was programmed in a way that it splits words by hyphens. Therefore we commented out a regex that does it and used a new tokenizer for better results. For example, the modified tokenizer successfully extracted ingredients such as lemon-pepper seasoning, all-purpose flour, etc. Also, thanks to its language-specific tokenization rules we could extract the whole ingredient description. To do this, initially, we got the root noun and then we joined all its compounds and constructed the full ingredient name (Algorithm 1). It is also necessary to omit the measurements stated in the third step of the pseudocode. The complete list of measurements is presented in Appendix A.

Algorithm 1 Constructing Ingredient Descriptions

```

1: Inputs: DataFrame df containing ingredient descriptions
2: Outputs: List of constructed ingredients
3: Define regular expression: measurements  $\leftarrow$ 
   bowl|bulb|cube|cup|pound|bite-size)s?}
4: for each row in df do
5:   Tokenize ingredient description: tokens  $\leftarrow$ 
     nlp(row['RecipeIngredient'])
6:   Initialize empty list: ingredient_parts  $\leftarrow$  []
7:   for each token in tokens do
8:     if token.dep_  $\in$  ['npadvmod', 'appos', 'pobj', 'dobj',
       'nsubj', 'ROOT']
       and token.pos_  $\in$  ['NOUN',
       'PROPN']
       and not measurements.match(token.text) then
9:       Add token.text to ingredient_parts
10:      for each child in token.children do
11:        if not measurements.match(child.text)
          and child.dep_  $\in$  ['compound',
          'amod', 'nmod'] then
12:          Append child.text to ingredient_parts
13:        end if
14:      end for
15:    end if
16:  end for
17:  Construct full ingredient list: full_description  $\leftarrow$ 
    join(ingredient_parts)
18: end for

```

At the end, by applying the mentioned technique it was possible to extract 7651 unique ingredients. In Table 1 the most common ingredients and their quantities are described.

Recipe Ingredient	Count
salt	4635
white sugar	3301
butter	3280
all-purpose flour	2632
eggs	1894
water	1837
olive oil	1751
onion	1742
milk	1580
vanilla extract	1570
brown sugar	1303
ground black pepper	1263
ground cinnamon	1157

TABLE I

RECIPE INGREDIENTS AND THEIR COUNTS

Additionally, below is an example of two recipes with their ingredients given in their prior format, and Table 2 presents the list of ingredients retrieved using the spaCy NLP tokenizer.

RECIPE 1 INGREDIENTS

Air Fryer Lemon Pepper Roasted Cashews:

- 1 cup raw cashews
- 1 tablespoon olive oil
- 0.5 teaspoon lemon-pepper seasoning
- 0.5 teaspoon garlic powder
- 0.5 teaspoon salt

RECIPE 2 INGREDIENTS

Quick Air Fryer Tilapia:

- olive oil cooking spray
- 2 tablespoons all-purpose flour
- 1.5 teaspoons seafood seasoning (such as Old Bay®)
- 0.75 teaspoon paprika
- 0.5 teaspoon garlic powder
- salt and freshly ground black pepper to taste
- 1 pound tilapia fillets, thawed if frozen
- 4 lemon slices

Recipe 1	Recipe 2
raw cashews	cooking spray
olive oil	all-purpose flour
lemon-pepper seasoning	seafood seasoning Bay ®
garlic powder	paprika
salt	garlic powder
	salt
	tilapia fillets

TABLE II

EXTRACTED INGREDIENTS

We can see from the previous table that for Recipe 1 all ingredients were correctly extracted, however, for Recipe 2 only "black pepper" and "lemon slices" could not be extracted by the tokenizer. Given that this model is statistically trained with an accuracy of approximately 97%, some errors can be expected.

After all the recipes have the ingredients formatted, and we know the recommended and non-recommended ingredients for each dosha we can continue with the calculation of the vata, pitta, and kapha score. Each dosha type's "favor" and "avoid" ingredient lists are obtained from a Dosha type Excel file.

So, we updated our dataset by adding additional columns to hold calculated scores for each dosha type. The score calculation is straightforward: for each recipe in the cleaned dataset, we look at the *recipe_ingredients* column value and check every ingredient used in that recipe; if it's in the "favor ingredient" list, we add one to the score; if it's in the "avoid ingredient" list, we subtract one. We apply the same method to every kind of dosha. We ignore the fact that ingredients are used in different proportions in recipes for easier calculation.

2) *Application of KNN Algorithm:* This healthy food recommendation application uses the K-Nearest Neighbors (KNN) algorithm to suggest recipes personalized to each user's dosha profile. KNN is a supervised learning technique that classifies data points based on their proximity to others

within a multidimensional feature space. This approach is well-suited for recipe recommendation as it returns a ranked list of recipes based on their similarity to a user's query. The algorithm was trained on a validated recipe database including various characteristics, such as macronutrient content, preparation time, and dosha type scores. This enables the system to identify similar recipes based on a user's initial selection, considering factors such as nutritional profile, time constraints, and Ayurvedic dietary principles. Consequently, the main goal implementing KNN was to facilitate efficient recipe exploration by providing precise and relevant suggestions for each recipe.

While the KNN algorithm significantly enhances the personalization of recipe recommendations by aligning them with users' nutritional preferences and dosha profiles, its application as a recommendation engine in this context presents two main challenges due to the nature of the available data set. Firstly, the data set suffers from a high degree of imbalance. This imbalance could potentially skew the recommendations, favoring recipes associated with more prevalent dosha profiles. When recommending recipes from under-represented doshas, their nearest neighbors (based on KNN) might often belong to different categories. To address this, two approaches were considered:

1. Penalty factors for different doshas: Introduce penalties during KNN calculations if a neighbor belongs to a different dosha type. This would downplay their influence on recommendations. We consider this approach as a better strategy for mitigating the data imbalance and deserves further exploration.

2. Increased neighbors and filtering: Increase the number of neighbors retrieved by KNN, then filter the results to only consider recipes from the same dosha category. However, this approach was discarded because it yielded less similar recipes for minority doshas when increasing neighbors.

Secondly, an analysis over the available data set, reveals that exists a minimal variation in the macronutrient profile and preparation time across recipes categorized under different doshas. This homogeneity within the data set limits the algorithm's ability to differentiate recommendations based on these dosha type criteria. To address this problem, a Principal Component Analysis (PCA) was employed to reduce the dimensionality of the feature space representing each recipe. PCA transforms correlated variables into a new set of uncorrelated principal components, ordered by the amount of variance they explain. This technique allows us to reduce the number of characteristics describing a recipe without significant information loss. By applying PCA, we were able to visualize the recipe set in two dimensions, revealing that recipes belonging to different doshas do not exhibit clear separation within this data set.

In the following Results and Discussion section, we provide a description of the data set on which the KNN algorithm has been fitted, an evaluation of the algorithm and we propose

a short discussion on the results obtained regarding the recommendation system implemented.

C. Scheduling of recipes

To optimize the scheduling of recipes, we employed a rigorous methodology to recommend only the most nutritionally suitable recipes to customers. The scoring criteria was based on essential nutritional characteristics, including the intake of carbohydrates, proteins, and sugars, aligning with dietary guidelines and recommended daily allowances [4].

By systematically exploring the recipes, we analyzed their nutritional components and provided a nutritional score for each. This score was based on a weighted analysis of the nutritional components, ensuring a balanced intake of nutrients and other dietary elements. The final recommendations were made based on the highest healthiness scores, ensuring that the selected meals contribute positively to the dietary needs and preferences of the Dosha Type of our customers. This structured, data-driven approach enhances the nutritional quality of the meal plans and promotes adherence to healthy eating habits by providing varied and appealing meal options. Through this methodology, we provide customers with the best possible dietary recommendations, contributing to their overall health and well-being.

III. RESULTS

In this section, we present the results obtained from the KNN model applied for recipe recommendation in the context of this application. As mentioned, the main objective of the model is to provide recipe suggestions that are similar in terms of nutritional profile and dosha affinity, thereby improving the user experience by offering relevant and personalized options. Two important aspects of a recommendation model have been evaluated, that is, the accuracy of the recommendations, and the consistency of the recommended recipes in relation to macronutrients, preparation time, and the score of belonging to a specific dosha. These results demonstrate the effectiveness of the KNN model in providing accurate and useful recommendations for users, aligned with the objectives pursued by this application.

Firstly, a table is presented below that shows the percentage of recipes belonging to each dosha-type.

Dosha	N° Recipes	Freq.
Vata	17,435	62%
Pitta	928	3%
Kapha	1,149	4%
No Dosha	8,579	31%

TABLE III
DISTRIBUTION OF RECIPES BY DOSHA TYPE

From this distribution it is possible to verify the class imbalance of the extracted recipes, which has been previously commented. Additionally, we wanted to measure the capacity of the implemented algorithm to recommend recipes to the user that should be consumed according to their dosha-type.

The following metric was defined: The average percentage of neighboring recipes of the same category for each dosha.

Dosha	Percentage
Vata	62.1%
Pitta	3.3%
Kapha	5.1%

TABLE IV

AVERAGE PERCENTAGE OF NEIGHBORS WITH THE SAME DOSHA TYPE

From this table, it is possible to conclude that the recipes recommended by the algorithm for Pitta and Kapha types are rarely of the same dosha. This is because, being less frequent in the dataset, it is also less likely that they will be found among the nearest neighbors.

Finally, two vectors of principal components were calculated in order to reduce the number of characteristics of each recipe and be able to observe how different the recipes collected are in terms of macro nutrients and preparation times, in order to verify if there is a significant difference. between each dosha.

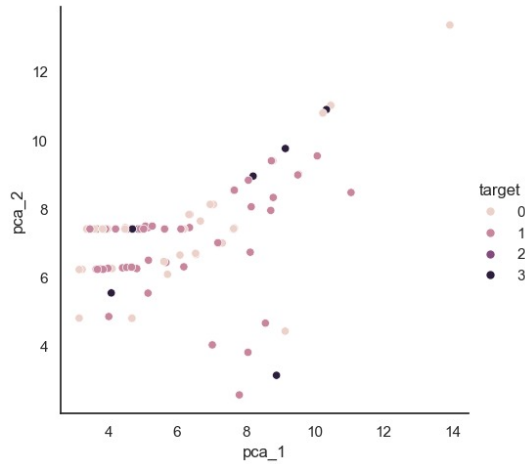


Fig. 1. PCA on complete data set

From this analysis of main components it is clear to see that there is no group of recipes belonging to a certain dosha that can be clearly differentiated from other types of recipes at least in terms of macro nutrients and preparation time. This can be seen graphically as each of these categories are not clearly separable. It should be noted that the labels of each group are as follows: 0: No dosha, 1: Vata, 2: Pitta, 3: Kapha.

IV. DISCUSSION

The current state of the implemented KNN algorithm indicates the need for several adjustments to enhance its ability to recommend relevant recipes to users. One of the primary challenges facing this implementation is the imbalance in the dataset, particularly with certain dosha

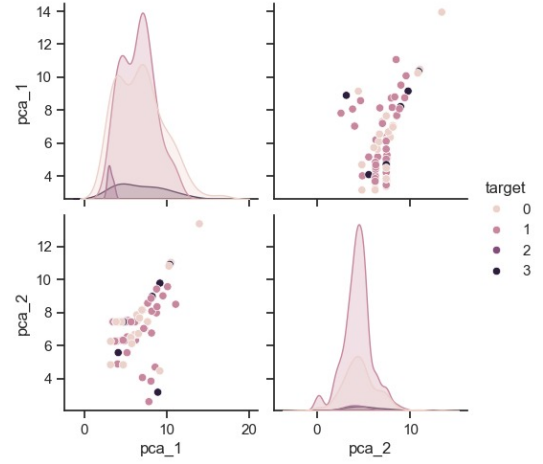


Fig. 2. Distribution of recipes by component

types (Pitta and Kapha) being underrepresented. This imbalance significantly impacts the algorithm's consistency in recommending recipes of the same dosha type to users. To address this issue, it is essential to augment the dataset by incorporating more recipes from the underrepresented dosha types. Additionally, it is crucial to introduce penalties when recommending recipes of different dosha types, effectively treating recipes that are similar but belong to a different category (due to their ingredients) as less similar. Furthermore, it is advisable to evaluate recipe similarity not only based on dosha type, preparation time and macronutrient content but also by considering other important factors like and recipe category and ingredients. The current approach of treating the entire dataset as a single group makes it challenging to accurately identify and recommend recipes that are genuinely similar in these specific aspects. For instance, evaluating macronutrients in a salad recipe versus a soup recipe can yield different results. Exploring the possibility of a KNN model that assesses similarity based on recipe composition and primary ingredients rather than just macro nutrients should also be considered.

By refining the criteria for similarity and addressing the dataset imbalance, significant improvements can be made to the accuracy and relevance of the recommendations.

V. DEVELOPMENT OF APPLICATION

A. Description

"AYURVEDA BASED RECIPE RECOMMENDATION" as an interactive web application, starting with the Welcome Page. This page features a dynamic video background with a central welcome message. Users are prompted to click the "Start" button to proceed to the next page.

The subsequent page presents users with 15 single-choice questions designed to collect user's information. Each question allows the user to select one answer from 3 options. As

users provide their responses, these are transmitted to the back-end API for processing. Users have the flexibility to navigate backward to modify their answers at any time. Upon completing all questions, the user clicks the "Finish" button to proceed.

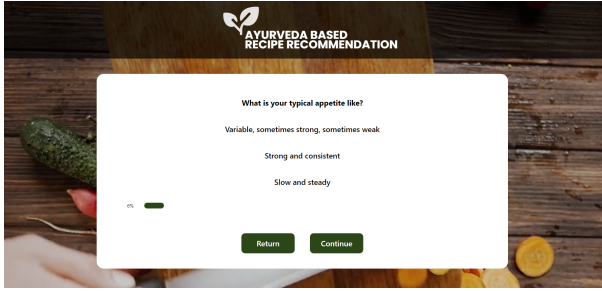


Fig. 3. Questionary Screen on the frontend

The user is then directed to a results page that displays their Ayurveda Dosha type based on the responses calculated by the back-end API. This page also provides a curated list of recommended recipes tailored to the user's Dosha type. Each recipe is displayed with its title, category, nutritional information (calories, protein content, fiber content, and fat content), and cook time. Additionally, an image corresponding to each recipe is shown. Users can click on the recipe name to navigate to a new page that includes detailed ingredients and cooking instructions. Furthermore, users have the option to click the "See More" button on the results page to view a list of similar recipes. Clicking on any recipe name within this list will provide more detailed information about that recipe.

Below the Dosha type display, there is a "See Your Weekly



Fig. 4. Questionary Result Screen

Planning" button. Clicking this button navigates the user to a

new page that shows a comprehensive weekly food planning. This planning outlines daily recipe recommendations for each of the seven days of the week, providing a appropriate dietary guide aligned with the user's Dosha type.



Fig. 5. Weekly Food Planning

B. Application Architecture

1) *Frontend and its specifications:* The architecture of the frontend application is meticulously designed to enhance efficient development practices. At the top level, the directory structure includes the 'src' folder, which serves as the primary source directory housing the core application logic. Within 'src', there is a subfolder named 'screens', further divided into individual folders for each screen component of the application and styling. Each screen's folder contains the necessary documents, code files, and resources specific to that screen's functionality. This modular structure enhances organization and facilitates streamlined management of the application's user interface components. Additionally, outside the 'src' folder, there exists an 'assets' directory dedicated to storing static resources such as images, videos, and other assets utilized throughout the application. By maintaining these resources within a separate folder, the project ensures a clean separation between source code and assets, which promotes easy access and manageability of resources.

This structured approach supports an efficient workflow and encourages collaboration among development team members by maintaining clarity and separation between core application logic and static resources.

The frontend architecture of the application is based on React.js, a widely used JavaScript library for building user interfaces. Complementing React.js, Vite is being used as a build tool, chosen for its fast bundling and efficient development workflow.

In our implementation, we integrate component libraries such as Chakra UI and React Bootstrap. These libraries offer pre-designed components and styling utilities, ensuring a consistent and responsive user interface design across our application. They streamline development by providing reusable UI elements that maintain visual coherence and usability.

For managing asynchronous operations and HTTP requests, we employ Axios. A library that simplifies data fetching and ensures reliable communication with the backend application, crucial for retrieving and processing data needed to recommend nutritionally optimal meal plans to the users.

This frontend architecture not only supports the application's functional requirements but also promotes scalability, maintainability, and user experience through efficient use of React.js and associated tools and libraries.

2) *Backend and its specifications*: The back-end system that supports the application has been developed using technologies that make the results of the recommendation model visible to users through its different views. Python has been used as the main programming language and Flask as a web framework due to its flexibility and simplicity in building web applications. The management and manipulation of our database is carried out using SQLAlchemy, an ORM (Object-Relational Mapper) that facilitates interaction with the database in a safe and efficient way. The back-end system is connected to a PostgreSQL database in which all the recipes and the main data of the application have been stored, such as user information, the answers sent by them and the dosha result calculated by answering the main questionnaire of the application. The questionnaire questions have also been stored in this database, as well as the calculation rules for each dosha. The back-end has been built following a development pattern based on micro-services, which allows the predictive model to be developed and scaled in parallel to other important functionalities of the application, such as user management and sending questionnaires. This modular architecture facilitates system maintenance and scalability, ensuring that each component can evolve independently.

The recipe recommendation model has been fitted using the scikit-learn library, usually used for its advanced capabilities in the development of Machine Learning models. Once tuned, the KNN model is saved and integrated within the application to make real-time predictions. Recipe data and recommendations can be accessed through specific URL's in our API: for example, the `/recipes/id` route delivers in JSON format all the relevant data for a specific recipe, such as its macro nutrients, cooking time, specific preparation instructions, image, author, a health score, and the Ayurvedic ingredients contained. While the path `/recipes/knn/id` returns a list of 20 similar recipes using the previously adjusted model. This RESTful API structure allows easy integration with the front-end and other services, ensuring fast and efficient communication between the different components of the application. For the purposes

of documenting the progress made in the development of this project, documentation in Postman of the main functionalities of this back-end system is made available to the reader. [5]

VI. CONCLUSION

Currently, our web application generates a fixed weekly food plan based on the user's identified Dosha type. This approach provides users with dietary recommendations that align with their Ayurvedic profile, ensuring that the meal plans support their overall wellness.

However, looking ahead, we envision a more advanced and personalized system. In our ideal future design, users will have the ability to select their preferred recipes. The application will then utilize this information, along with their Dosha type, to intelligently generate a customized weekly or even monthly meal plan. This enhanced functionality aims to offer a more tailored and flexible dietary solution, accommodating individual tastes and dietary preferences while maintaining the Ayurvedic principles. Additionally, it is common for current Ayurveda Dosha type tests to yield mixed results, such as Vata-Pitta, Pitta-Kapha, or Vata-Kapha. Due to time constraints and the complexity involved, our current system provides users with their primary single Dosha type. In the future, we plan to refine this aspect to deliver more realistic results, reflecting the mixed Dosha types that are more accurate for many users.

Furthermore, our next step will be to refine the NLP technique to extract recipe components more precisely, as well as to standardize recipe categories in order to create a better weekly meal plan for the entire day.

Such an improvement would not only increase user satisfaction by allowing more control over their meal choices but also leverage intelligent algorithms to optimize meal planning for health and enjoyment. This represents a significant step forward in creating a more user-centric and adaptable dietary planning tool.

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APPENDIX

measurements = [bowl, bulb, cube, clove, cup, drop, ounce, oz, pinch, pound, quart, tablespoon, teaspoon, sprinkle, bite-size, fry-size, half, piece, inch, tbsp, tsp, g, kg, gram, kilogram, ml, liter, litre, milliliter, bite-size, pint-sized, container, slice, sliced, degree, large, extra-large, lengthwise, half, small, can, package, whole, miniature, jar, mini, medium, sticks, bottle, bottled, box, head, bottle, loaf, round, thickcut, part, diced, additional, toothpicks, carton, lunch, portion, dry, dry-curd, curd, sprig, shredded, lowfat, low-fat, fat-free, creamed, 1, thick, reduced-fat, nonfat, fat, light, packed, reduced fat, pint, unsweetened, high-quality, stick, unsalted]